

PHARMA ANALYTICS POWERBI PROJECT ARCHITECTURE DOCUMENT

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TOOL : Microsoft powerbi

DATABASE : Snowflake

PROJECT : Pharma analytics dashboard

PROJECT ARCHITECTURE OVERVIEW

The Pharma Analytics Power BI project is designed to analyze pharmaceutical sales, product performance, profitability, quantity, and regional business performance. The data is stored and managed in Snowflake and connected to Power BI for data modeling, DAX calculations, visualization, analysis, and reporting.

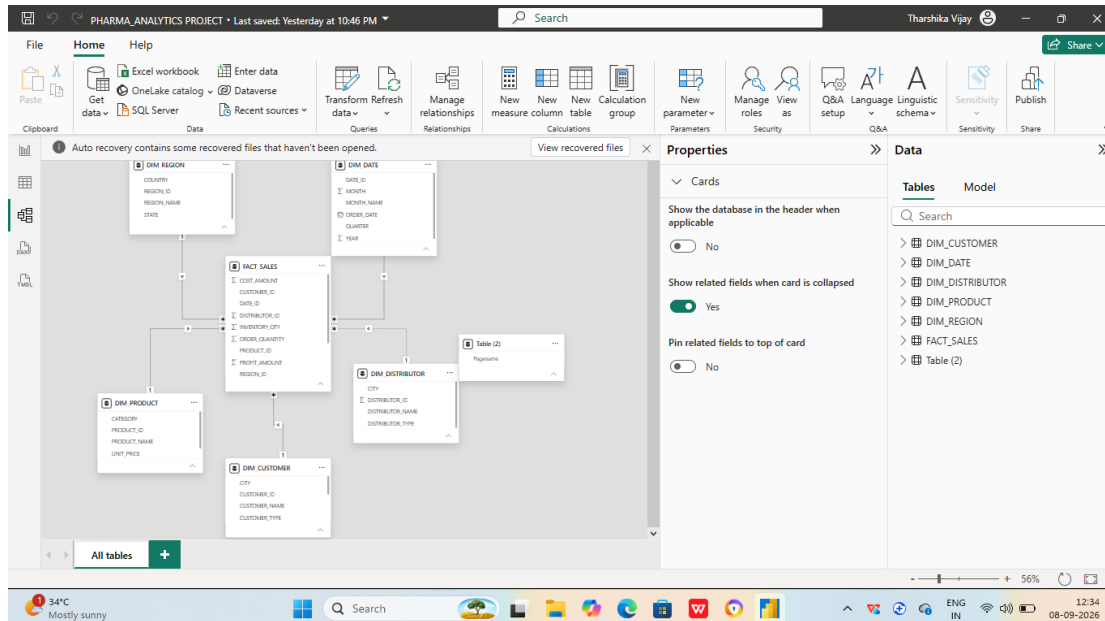
DATA FLOW

[Snowflake Database] → [Data Model / Star Schema] → [DAX & Calculations] → [Power BI Report] → [Power BI Service]

DATA MODEL ARCHITECTURE

The Pharma Analytics project follows a Star Schema architecture. FACT_SALES is the central fact table, connected to multiple dimension tables through one-to-many relationships. The dimension tables provide descriptive information for analyzing sales, products, customers, distributors, dates, and regions.

DATA MODEL DIAGRAM



TABLES AND COLUMNS

FACT_SALES – Fact Table

- COST_AMOUNT
- CUSTOMER_ID
- DATE_ID
- DISTRIBUTOR_ID
- INVENTORY_QTY
- ORDER_QUANTITY
- PRODUCT_ID
- PROFIT_AMOUNT
- REGION_ID

DIM_PRODUCT – Dimension Table

- PRODUCT_ID
- PRODUCT_NAME
- CATEGORY
- UNIT_PRICE

DIM_CUSTOMER – Dimension Table

- CUSTOMER_ID
- CUSTOMER_NAME

- CUSTOMER_TYPE
- CITY

DIM_DISTRIBUTOR – Dimension Table

- DISTRIBUTOR_ID
- DISTRIBUTOR_NAME
- DISTRIBUTOR_TYPE
- CITY

DIM_REGION – Dimension Table

- REGION_ID
- REGION_NAME
- STATE
- COUNTRY

DIM_DATE – Dimension Table

- DATE_ID
- ORDER_DATE
- MONTH
- MONTH_NAME
- QUARTER
- YEAR

Table (2) – Helper Table

- PageName

RELATIONSHIPS

The dimension tables are connected to FACT_SALES using one-to-many (1:*) relationships. The dimension tables act as the “one” side and FACT_SALES acts as the “many” side.

- DIM_PRODUCT[PRODUCT_ID] → FACT_SALES[PRODUCT_ID]
- DIM_CUSTOMER[CUSTOMER_ID] → FACT_SALES[CUSTOMER_ID]
- DIM_DISTRIBUTOR[DISTRIBUTOR_ID] → FACT_SALES[DISTRIBUTOR_ID]
- DIM_REGION[REGION_ID] → FACT_SALES[REGION_ID]
- DIM_DATE[DATE_ID] → FACT_SALES[DATE_ID]

DATA FLOW ARCHITECTURE

SNOWFLAKE DATABASE



FACT & DIMENSION TABLES



POWER BI DATA MODEL (STAR SCHEMA)



DAX MEASURES & CALCULATIONS



POWER BI REPORT / DASHBOARD



POWER BI SERVICE

Snowflake is used as the backend data platform for storing the pharmaceutical analytics data. The Fact and Dimension tables are connected to Power BI and organized using a Star Schema. DAX measures and calculations are then applied to create business insights and interactive visualizations. The completed report is published to Power BI Service for user access and analysis.